

- (c) Mediterranean Sea
(d) Atlantic Ocean

M.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of the above question.

5. The “Great Barrier Reef” of corals is situated near the coast of:

- (a) New Zealand (b) Finland
(c) Ireland (d) Queensland

R.A.S./R.T.S. (Pre) 1993

Ans. (d)

See the explanation of the above question.

6. The largest Coral reef in the world is found near the coast of which one of the following Countries?

- (a) Australia (b) Cuba
(c) Ghana (d) Philippines

I.A.S. (Pre) 2007

Ans. (a)

See the explanation of the above question.

7. Consider the following statements :

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and Philippines.
3. Coral reefs host far more number of animal phyla than those hosted by tropical rainforests.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (d)

Coral reefs are formed by the consolidation and combination of the skeletons of marine organisms, coral or coral polyps. Significantly, coral reefs generally are confined to tropical and semi-tropical waters (mostly 25°N to 25°S). Somewhere their expansion is 30°N to 30°S. Hence statement (1) is correct.

According to Reef Base : A Global Information System for Coral Reef, about 19 % of the world's total coral reef areas are in Australia and 17% in Indonesia respectively. Thus it can be said that more than one-third of the world's coral reefs are found in the territories of Australia, Indonesia and

the Philippines. Hence statement (2) is also correct. Some of the renowned environmentalists estimate that more than 25,000 described species from thirty two of the world's thirty-three animal phyla live in reef habitats. It is almost four times the number of animal phyla found in tropical rainforests. Thus we can say that statement (3) is also correct. Hence option (d) is correct.

8. Addu Atoll are situated in which ocean?

- (a) Atlantic Ocean
(b) Arctic Ocean
(c) Indian Ocean
(d) Pacific Ocean
(e) None of the above/More than one of the above

66th B.P.S.C. (Re-exam) (Pre) 2020

Ans. (c)

Addu Atoll is situated in the Indian Ocean. It is the southernmost atoll of the Maldives.

9. The shape of Atoll is :

- (a) Rectangular
(b) Rod-shaped
(c) Horse Shoe/Ring shaped
(d) Triangular

Uttarakhand P.C.S. (Pre) 2024

Ans. (c)

An atoll is a ring-shaped island that includes a coral reef surrounding a lagoon. They are generally circular or horseshoe-shaped, with diameters that can span several kilometers. These structures are typically found in tropical and subtropical oceans and seas where coral can grow. Atolls can be complex and vibrant structures. They may include various reefs that are large enough to support tropical islands and even small towns.

The Atmosphere

*The atmosphere is made up of different gases. Among all these gases oxygen and carbon dioxide are important for the existence of humans and plants respectively. * The atmosphere is composed of various gases, of which 99% is made up of nitrogen and oxygen. Other 1% comprises argon, carbon dioxide, neon, krypton, helium, Hydrogen, and xenon.

Gas	Formula	Presence in Atmosphere (in percentage)
Nitrogen	N ₂	78.084
Oxygen	O ₂	20.947
Argon	Ar	0.934
Carbon dioxide	CO ₂	0.0407
Neon	Ne	0.0018
Helium	He	0.0005
Krypton	Kr	0.0001
Hydrogen	H ₂	0.000053
Xenon	Xe	0.000009

*Heavy gases in the atmosphere descend whereas lighter gases are found in the upper layers as they ascend. As a result CO₂ gases extend up to 20 km from the surface, oxygen, and nitrogen up to 100 km, hydrogen up to 125 km and other gases rise far above 125 km except ozone. The atmosphere is made up of different layers of gases with varying densities and temperatures. Based on these factors different layers of the atmosphere are as follows-

(1) Troposphere – The lowest layer of the atmosphere is called troposphere. All weather activities like clouds, hail, fog, blizzards, thunder, windstorms, etc. are confined to the troposphere. The most important characteristic of the troposphere is that there is a decrease of **6.5° C** with an increase of every 1000 m in altitude. This decrease in temperature with an increase in altitude is known as normal lapse rate. The upper boundary of the troposphere is called tropopause.

(2) Stratosphere – Stratosphere (15-50 km) is the second most important layer of the atmosphere. It lies between mesosphere (above) and troposphere (below). This layer of atmosphere is very dry in nature and has the least amount of water vapor in it. As a result, there is no weather activity in stratosphere. The stratosphere is used for **aviation purposes** as the absence of weather activities makes this layer ideal for flying. Ozone layer is found in this sphere.

(3) Mesosphere – This layer is found at an altitude of 50 to 80 km. The upper ceiling of mesosphere is called mesopause.

(4) Ionosphere – It is an electrically conducting layer, that makes **radio wave transmission** possible over long distances. Ionosphere extends from 80 to 640 km. *The unique feature of Ionosphere is the phenomenon of **Aurora Borealis** in Northern Hemisphere and **Aurora Australis** in Southern Hemisphere.

(5) Exosphere – It is the topmost layer of the atmosphere and

is situated above the ionosphere. This layer has **no defined boundary**. The **Communication satellites** are positioned in the exosphere.

1. Consider the following statements:

Statement-I : Thickness of the troposphere at the equator is much greater as compared to poles.

Statement-II : At the equator, heat is transported to great heights by strong convectional currents.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
- (b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
- (c) Statement-I is correct, but Statement-II is incorrect
- (d) Statement-I is incorrect, but Statement-II is correct

Ans. (a)

I.A.S. (Pre) 2024

The troposphere is the lowermost layer of the atmosphere. This layer contains dust particles and water vapor. All changes in climate and weather take place in this layer. The temperature in this layer decreases at the rate of 1°C for every approximately 165 m of height. This is the most important layer for all biological activity. Its average height is about 13 km and extends roughly to a height of about 8 km near the poles and about 18 km at the equator. The thickness of the troposphere is greatest at the equator because at the equator, heat is transported to great heights by strong convectional currents. Hence, both Statements-I and Statements- II are correct and Statements-II explains Statement-I.

2. The atmosphere is a mixture of several gases. Near the earth surface it contains mainly:

- (a) Nitrogen and oxygen
- (b) Nitrogen and carbon dioxide
- (c) Oxygen and carbon dioxide
- (d) Ethane and oxygen

R.A.S./R.T.S. (Pre) 1997

Ans. (a)

The atmosphere is a mixture of different gases, of which 99% are mainly nitrogen (78.084%) and oxygen(20.947%). Remaining approximately 1% consists of argon, carbon dioxide, neon, helium, methane, hydrogen, nitrous oxide, ozone, krypton, xenon, etc. Heavy gases are found at the bottom of atmosphere and light gases are found at the top. Carbon dioxide is found up to 20 km above Earth, oxygen and nitrogen up to 100 km, hydrogen up to 125 km and except for ozone other gases are found at much higher elevations.

3. Which gas has a maximum percentage in the atmosphere?

- (a) Carbon (b) Nitrogen
(c) Oxygen (d) Hydrogen

U.P.P.C.S. (Pre) 1993

Ans. (b)

See the explanation of the above question.

4. Which one of the following is the permanent gas in the atmosphere?

- (a) Carbon dioxide (b) Ozone
(c) Nitrogen (d) Neon

U.P. P.C.S. (Mains) 2017

Ans. (c)

Permanent gases in the atmosphere are those gases whose quantity does not change from day to day. These gases primarily contain nitrogen, oxygen and argon. Except for these three gases, the quantity of CO₂ ozone and neon keeps far changing according to the circumstances.

5. The correct sequence of different layers of the atmosphere from the surface of the Earth upwards is :

- (a) Troposphere, Stratosphere, Ionosphere, Mesosphere
(b) Stratosphere, Troposphere, Ionosphere, Mesosphere
(c) Troposphere, Stratosphere, Mesosphere, Ionosphere
(d) Stratosphere, Troposphere, Mesosphere, Ionosphere

I.A.S. (Pre) 1998

Ans. (c)

On the basis of temperature and air pressure atmosphere is divided into five layers starting from the Earth's surface which are as follows :

Troposphere- (15 km from the surface of the Earth)

Stratosphere (15-50 km)

Mesosphere (50-80 km)

Ionosphere (80-640 km)

Exosphere (above more than 640 km)

The thermosphere is located above the mesosphere and below the exosphere, and its lowest part contains ionosphere.

So option (c) is correct.

6. In the vertical structure of the atmosphere, which one is the lowest layer?

- (a) Stratosphere
(b) Mesosphere
(c) Troposphere
(d) Thermosphere

U.P. R.O./A.R.O. (Re Exam) (Pre) 2016

Ans. (c)

In the vertical structure of the atmosphere, Troposphere is the lowermost layer. Its average elevation is about 12 km. All seasonal phenomena - like rain, fog, hail, etc. occur on this surface.

7. Which of the following statements is true about troposphere?

- (a) Its average height is 13 km.
(b) It is the topmost layer of the atmosphere.
(c) The temperature at this layer increases with the height.
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

See the explanation of the above question.

8. The four layers of the atmosphere are –

1. Ionosphere 2. Mesosphere
3. Stratosphere 4. Troposphere

Their correct ascending order in terms of height is

Code :

- (a) 1, 2, 3, 4 (b) 2, 1, 4, 3
(c) 4, 3, 2, 1 (d) 3, 4, 1, 2

U.P.P.C.S. (Mains) 2005

Ans. (c)

See the explanation of the above question.

9. Arrange the following atmospheric zones in terms of the distance from the earth's surface in increasing order and select the correct answer from the codes given below:

1. Thermosphere 2. Troposphere
3. Stratosphere 4. Mesosphere

Code :

- (a) 1, 3, 4, 2 (b) 2, 3, 1, 4
(c) 3, 4, 2, 1 (d) 2, 3, 4, 1

U.P.P.C.S. (Pre) (Re-Exam) 2015

Ans. (d)

See the explanation of the above question.

10. Most weather activity occurs in which atmospheric layer:

- (a) Ozonosphere
(b) Ionosphere
(c) Troposphere
(d) Exosphere

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

Almost all the weather phenomena like rainfall, fog and hailstorms, etc. occur in the troposphere. The temperature in this layer decreases at the rate of 6.5°C for every 1000 meters (1 km) in height. This is the most important layer for all biological activity.

11. Which of the following atmospheric layers is known for all changes in climate and weather?

- (a) Thermosphere
- (b) Stratosphere
- (c) Mesosphere
- (d) Troposphere

M.P.P.C.S. (Pre) 2023

Ans. (d)

See the explanation of the above question.

12. The stratosphere is said to be ideal for flying jet aircraft. This is because-

- (a) This layer is rich in ozone which reduces fuel consumption
- (b) The temperature is constant and ideal for aircraft engine efficiency
- (c) This layer is out of the firing range of antiaircraft guns
- (d) The absence of clouds and other weather phenomena

Jharkhand P.C.S. (Pre) 2010

Ans. (d)

The stratosphere is the second major layer of Earth's atmosphere, just above the troposphere, and below the mesosphere. The stratosphere is very dry and the air in this layer contains little water vapor. Because of this, stratosphere is almost free from clouds and associated weather phenomena. These conditions are ideal for avoiding disturbances. Jet aircraft use this layer to fly.

13. The ozone layer lies in the :

- (a) Troposphere
- (b) Tropopause
- (c) Stratosphere
- (d) Photosphere

Chhattisgarh P.C.S. (Pre) 2011

Uttarakhand P.C.S. (Pre) 2010

U.P.P.C.S. (Pre) 1997

Ans. (c)

The ozone layer is mainly found in the lower portion of the stratosphere from 15 to 35 kilometers above Earth. The maximum height of this layer is 55 kilometers but the maximum concentration of ozone layer is found from 15 to 35 kilometers. This layer absorbs ultra-violet radiation and shields life on the Earth from an intense and harmful form of energy. Ultra-violet rays may cause skin cancer.

14. The Ozone layer lies in the –

- (a) Troposphere
- (b) Mesosphere
- (c) Stratosphere
- (d) Ionosphere

U.P.P.C.S. (Mains) 2005

U.P.P.C.S. (Mains) 2003

Ans. (c)

See the explanation of the above question.

15. Which of the following protects life on earth from the harmful radiations of the Sun ?

- (a) Troposphere
- (b) Ionosphere
- (c) Ozone layer
- (d) Mist
- (e) All these.

Chhattisgarh P.C.S (Pre) 2013

Ans. (c)

The ozone layer protects us from the harmful ultra-violet radiations of the Sun. In the atmosphere, it is mainly found in the lower portion of the stratosphere from 15–35 km above Earth. Ozone is a molecule composed of three oxygen atoms. Its colour is blue.

16. The function of the ozone layer in the stratosphere is –

- (a) To stabilize the global temperature
- (b) To reduce the frequency of earthquake
- (c) To avoid failure of monsoons
- (d) To prevent ultra-violet radiation effect on the ground

Uttarakhand P.C.S. (Pre) 2002

Ans. (d)

See the explanation of the above question.

17. Ozone layer refers to –

- (a) The atmospheric condition of Antarctica
- (b) Modern invention done on the planet Saturn
- (c) The layer about 10-20 km. below the surface of Earth
- (d) The layer of atmosphere of 15-20 km. above the surface of Earth

38th B.P.S.C. (Pre) 1992

Ans. (d)

See the explanation of the above question.

18. The height of the ozone layer above the surface of the earth is-

- (a) 15-20 km
- (b) 40-50 km

- (c) 70-80 km (d) 110-120 km

41st B.P.S.C. (Pre) 1996

Ans. (a)

See the explanation of the above question.

19. Which one of the following layers of the atmosphere is responsible for the deflection of radio waves?

- (a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Ionosphere

U.P.P.C.S. (Mains) 2005

I.A.S. (Pre) 1996

Ans. (d)

Radio waves are reflected by the Ionosphere. Other major events that occur in this region are as follows- Identification of cosmic rays and observation of Aurora Borealis and Aurora Australis.

20. Which layer of atmosphere is responsible for Aurora Borealis?

- (a) Troposphere (b) Thermosphere
(c) Ionosphere (d) Exosphere

U.P.P.C.S (Pre) 2010

Ans. (c)

Aurora is an incredible light caused by collisions between electrically charged particles released from the Sun that enter the Earth's atmosphere. Aurora that occurs in the Northern Hemisphere is called 'Aurora Borealis' and in Southern Hemisphere is called 'Aurora Australis'. The light of aurora extends from (80 km to 640 km). The ionosphere is responsible for the Aurora Borealis.

21. Wireless communication is reflected to the earth's surface by –

- (a) Troposphere
(b) Stratosphere
(c) Ionosphere
(d) Exosphere

U.P.P.C.S.(Pre) 2013

Ans. (c)

The Ionosphere is composed of electrically charged particles that are called 'ions'. These particles reflect radio waves back to the Earth's surface and enable us to have wireless communication.

22. Which layer of Atmosphere is used for Telecommunication?

- (a) Troposphere (b) Thermosphere
(c) Stratosphere (d) Ionosphere

Jharkhand P.C.S. (Pre) 2013

Ans. (d)

See the explanation of the above question.

23. In which atmospheric layer are communication satellites located?

- (a) Exosphere
(b) Stratosphere
(c) Ionosphere
(d) Troposphere

Uttarakhand P.C.S. (Pre) 2005

R.A.S./R.T.S. (Pre) 1997

Ans. (a)

Geostationary satellites are positioned at approximately 36000 km height above the Earth. The exosphere is the outer limit of the atmosphere above 640 km from the Earth. Therefore, the communication satellites located in the outermost layer of the atmosphere are in the exosphere.

24. Which of the following are correct in matching the layer of atmosphere and its characteristics ?

1. Troposphere – Weather Phenomena
2. Stratosphere – Ozone layer
3. Ionosphere – Radio waves reflected to Earth's surface
4. Mesosphere – Aurorae

Code :

- (a) 1, 2, 3 and 4 (b) 1, 2 and 4
(c) 1, 2, and 3 (d) 2 and 3

U.P.P.C.S. (Mains) 2002

Ans. (c)

Almost all the weather phenomena like rainfall, fog and hailstorm etc. occur in the troposphere.

Stratosphere - contains a layer of ozone gas.

Radio waves transmitted from the Earth are reflected back to the Earth by ionosphere. Auroras are also related to this layer. So, option 1,2 and 3 are correctly matched, while options (4) is incorrect.